



Analyzing volatility transmission using group transfer entropy

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ABSTRACT

We analyze the transmission of volatility between the oil, stock, gold, and currency markets using transfer entropy. Our approach, which we denote by group transfer entropy, measures the overall sensitivity of a process to lagged realizations of a group of other processes. We supplement our measure by a suitable block bootstrap approach which allows to conduct inference. Our empirical analysis uses daily data of OVX, VIX, GVZ, and EVZ from 2008 to 2017. The results show that oil and stock market volatility are the most affected by past volatility changes in the other markets, which highlights the importance for investors to hedge against this predictable risk transmission. Furthermore, we show that group transfer entropy can unveil nonlinear dynamics beyond the standard vector autoregressive methodology and provides a useful basis for forecast averaging.

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1. Introduction

Following the financialization and increasing integration of commodity markets, more and more investors engage in global trading strategies, adding commodities, such as oil or gold, to their portfolios to achieve optimal risk diversification. During the global financial crisis in 2007/2008, however, contagion led to increasing uncertainty across a wide range of markets, revealing the importance of a better understanding of volatility transmission mechanisms (Creti et al., 2013; Henderson et al., 2015).

On the one hand, equity market volatility might be considered as the most general indicator of uncertainty, as stock and option prices tend to react very fast to changing external conditions. On the other hand, commodity prices, and in particular oil prices, are more prone to exogenous shocks which stem from increased uncertainty due to the geopolitical and global economic framework. Such shocks might increase oil price volatility, which in turn could affect exchange

rate and stock market volatility when uncertainty concerning economic prospects increases (Liu et al., 2013). Bi-directional volatility spillovers between the oil and specific stock market sectors have already been documented, for instance in Malik and Ewing (2009) and Arouri et al. (2011). As outlined by Hammoudeh et al. (2010), oil prices also have an impact on the sensitivity of volatility spillovers between precious metals and exchange rate volatility. These results suggest that volatility transmission mechanisms include more than two markets.

Implied volatility indices across different asset classes allow to investigate these volatility spillovers. They measure a market's expectation of 30-day volatility using data on options over a wide range of strike prices. Originally, the methodology was applied by the Chicago Board Options Exchange (CBOE) to measure volatility in the equity markets with the VIX (S&P 500 implied volatility index) as the most prominent example. However, following the increasing interest of market participants in this kind of volatility measures, the CBOE started to publish implied volatility indices for commodity markets as well, starting with the oil market, the Crude Oil Volatility Index (OVX), and the gold market, the CBOE Gold Volatility Index (GVZ), in 2007. A corresponding measure for foreign exchange rate volatility, the EVZ, was implemented in 2008.

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The number of studies which use implied volatility indices to examine spillovers across different markets has increased in recent years. Most closely related to our analysis is the study by Liu et al. (2013) who examine oil, gold, stock, and foreign exchange rate implied volatility indices. They find that oil market volatility is significantly influenced by the other market's volatility. However this influence seems to be short lived and transient.

It is not straightforward to model all direct and indirect channels through which volatility is transmitted across different asset classes at once. However, from the perspective of an investor seeking to hedge her portfolio, it would be of great interest to which extent an asset is affected by volatility spillovers from other markets, and whether a dominant direction of volatility spillovers exists. Our study seeks to shed some light on this issue. Our contribution to the current literature is therefore fourfold: First, we examine potential spillover effects by analyzing oil, stock, gold, and exchange rate volatility as measured by the CBOE's implied volatility indices (OVZ, VIX, GVZ, EVZ). We analyze daily data from June 2008 to May 2017, i.e. a period situated after the 2007/2008 financial crisis and consequently examine the interlinkages in a "normal" trading environment. Second, we analyze the transmission of volatility from a group of markets to each single market, as we are interested in measuring the extent to which each market's volatility is affected by a set of other markets. To this end, we rely on our implementation of transfer entropy to which we refer to as Group Transfer Entropy (GTE). Third, we complement the study by Liu et al. (2013), who measure volatility transmission between implied volatility indices of oil, stock, gold market, and the exchange rate, relying on a vector autoregressive (VAR) model. We show that a more general approach such as GTE can detect transmission processes which are not captured by such linear time series models. Fourth, we illustrate how GTE can be applied to find optimal weights in a forecast averaging context. We conclude that the GTE based forecast performs better than the VAR based forecast for all volatility indices and also outperforms an equal weighting scheme.

VAR models have so far been the workhorse when examining interdependencies between different markets (e.g. Badshah et al., 2013; Fernandez-Perez et al., 2017). Arouri et al. (2011) detect significant volatility spillovers between the oil and stock markets based on a VAR-GARCH model.¹ In our empirical analysis we show that volatility spillovers exist beyond those indicated by the standard VAR approach: we first fit a VAR model to purge the time series of linear spillovers, and subsequently use the residuals of this model as input for transfer entropy calculation. Our results reveal that there exist significant dependencies between the extracted residual series, implying that linear models only measure a subset of the spillovers inherent in the data.

Transfer entropy has been applied in previous studies to measure information flows between financial markets by Dimpfl and Peter (2013, 2014), Kwon and Yang (2008), and He and Shang (2017). Other applications include - but are not limited to - Chunxia et al. (2016), who study the contagion among American industries based on transfer entropy, or Teng and Shang (2017) who investigate transfer entropy for different time scales. We complement these studies by using a transfer entropy estimator suitable for our research question, which we call Group Transfer Entropy. GTE measures the dependence of the current realization in a process on past realizations of a group of other processes. Additionally, we use a suitable time series

block bootstrap, which enables us to conduct inference on the estimated GTE measures. Our results show that, based on our sample, there are only weak volatility spillovers when a VAR framework is used. However, applying GTE we detect significant spillovers, both in the original time series as well as in the residual processes of the VAR. We document that the largest volatility transmission is directed towards the oil market and the stock market, while gold and US dollar exchange rate volatilities are the least affected by past volatility in the other markets. This result contributes to the general understanding of volatility transmission and might be of particular interest for risk managers and investors seeking to optimally hedge their portfolio. Finally, we show how investors can use GTE to forecast volatility and that this approach is well suited to deliver forecasts including information beyond the standard VAR approach.

The article proceeds as follows. Section 2 introduces the GTE methodology and subsequently presents the data. Section 3 holds the empirical results. Section 4 presents the application of GTE to forecast averaging and Section 5 concludes.

2. Methodology and data

2.1. Group transfer entropy

Transfer entropy is a measure that captures the reduction of uncertainty when predicting a variable I by using additional information contained in process J . In economic terms, reduction of uncertainty means nothing but the flow of information. In the bivariate case, Schreiber (2000) proposes to measure the information flow from process J to process I by quantifying the deviation from the generalized Markov property $p(i_{t+1}|i_t^{(k)}) = p(i_{t+1}|i_t^{(k)}, j_t^{(l)})$, where $p(i_{t+1}|i_t^{(k)})$ denotes the conditional probability for observing i_{t+1} given its own history only, and $p(i_{t+1}|i_t^{(k)}, j_t^{(l)})$ is the conditional probability for observing i_{t+1} given its own history and the history of process J . k (l) denotes the order of the Markov process for I (J). In the case of no lagged dependencies between J and I , the transition probabilities are not affected by conditioning on previous observations of J and the equality holds. Transfer entropy as introduced by Schreiber (2000) measures the deviation of the bivariate system from this assumption based on the Kullback-Leibler distance

$$T_{J \rightarrow I}(k, l) = \sum p(i_{t+1}, i_t^{(k)}, j_t^{(l)}) \times \log \frac{p(i_{t+1}|i_t^{(k)}, j_t^{(l)})}{p(i_{t+1}|i_t^{(k)})}. \quad (1)$$

$T_{J \rightarrow I}$ thus measures the information flow from process J to process I . Note that transfer entropy is an asymmetric measure. This means that by exchanging J and I and calculating $T_{I \rightarrow J}(l, k)$ along with $T_{J \rightarrow I}(k, l)$ we can infer the dominant direction of the information transmission.² In our multivariate application, we consider four processes and seek to measure the information flow from a group of three variables to the remaining one. Let us assume we observe four processes I, J, X , and Y . We are interested in the information flow of the group of the processes J, X , and Y to the process I . Building

¹ Arouri et al. (2011) define a bivariate VAR(1)-GARCH(1,1) model using stock market and oil index returns. The GARCH (1,1) specification of the conditional variances thereby includes lagged squared innovations and first-order autoregressive conditional variance components of the stock as well as the oil market return series, which allow to analyze spillover effects.

² While a decomposition into the individual effects might be interesting, Papana et al. (2013) show by means of a simulation study that partial transfer entropy is often unable to detect the relationships or gives spurious results in small samples as in the present application. A sufficiently large sample would require more than 50,000 observations which is infeasible in economic applications which are to a large extent conducted on data of daily or even lower frequency.

on the idea of transfer entropy, we use a corresponding measure for information flow from a group of processes to a single process. The group transfer entropy $GTE_{(J,X,Y) \rightarrow I}(k,l)$ is given by the following formula:

$$GTE_{(J,X,Y) \rightarrow I}(k,l) = \sum p(i_{t+1}, i_t^{(k)}, j_t^{(l)}, x_t^{(l)}, y_t^{(l)}) \times \log \frac{p(i_{t+1} | i_t^{(k)}, j_t^{(l)}, x_t^{(l)}, y_t^{(l)})}{p(i_{t+1} | i_t^{(k)})}. \quad (2)$$

In order to implement the measures in Eqs. (1) or (2), the conditional probabilities are replaced by unconditional probabilities using Bayes' law and the probabilities are estimated by empirical frequencies.

The major advantage of using transfer entropy in general and the implementation as group transfer entropy in particular lies in the ability of transfer entropy to accommodate nonlinear relationships between time series. In an extreme case where there is no linear dependence at all, a VAR model might (wrongly) indicate that the time series are independent, while GTE reliably detects any form of dependence. Furthermore, the methodology is robust to rare outliers and structural breaks. The reason is that a single outlier or a once occurring structural break have only a negligible impact on the overall estimation of the empirical frequencies.³

While financial data are realizations of continuous processes, the calculation of GTE in Eq. (2) requires discrete input. Therefore, to implement the methodology, discretization of the data is required. Motivated by previous studies using financial data (cp. Dimpfl and Peter, 2013) we propose to apply a partitioning into three bins. A higher number of bins is possible, but requires a considerably greater amount of data. To obtain the discretized series $S(t)$ a stationary time series $x(t)$ is partitioned as follows:

$$S(t) = \begin{cases} A & \text{for } x(t) \leq q_1 \\ B & \text{for } q_1 < x(t) < q_2 \\ C & \text{for } x(t) \geq q_2 \end{cases}. \quad (3)$$

The symbolic encoding in Eq. (3) replaces each value in the observed time series by a corresponding symbol (A, B, C), according to how $S(t)$ relates to the interval specified by the quantiles q_1 and q_2 . The choice of the quantile should be motivated by the distribution of the data. In the case of financial data, tail observations are commonly of particular interest and rather low and high quantiles are chosen. In order to check for robustness of the transfer entropy measure, the chosen quantiles can be varied.

As transfer entropy suffers from a positive small sample bias, we calculate effective group transfer entropy (EGTE) by subtracting the bias which is estimated from shuffling. In particular, we keep the dependent variable I to conserve its autoregressive properties, but draw random triples (j_t, x_t, y_t) from the group of variables J, X , and Y in order to fully destroy any existing dependencies with I . Hence, the inclusion of the shuffled group should not be any more helpful in predicting I than a randomly drawn i.i.d. sequence. In the limit, the so estimated shuffled GTE converges to zero. Re-estimation of GTE using the shuffled series yields an estimate of the bias which is subsequently subtracted from the estimate in Eq. (2). We repeat the shuffling 100

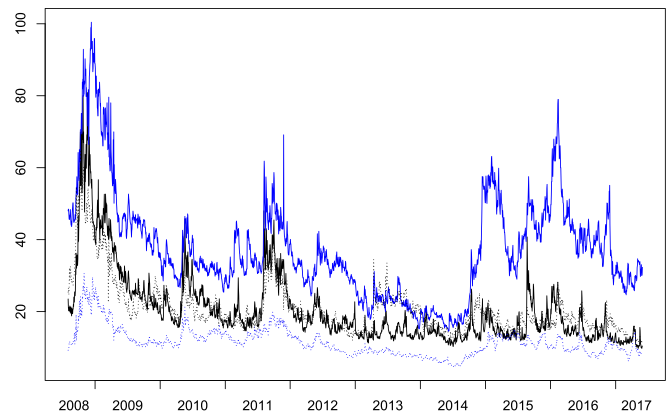


Fig. 1. Time series plots. The figure presents time series plots for OVX (blue solid), VIX (black solid), GVZ (black dotted), and EVZ (blue dotted) from June 3, 2008 to May 31, 2017.

times and use the average over these repetitions as an estimator for the bias.

To derive the distribution of the EGTE estimator, we rely on a time series block bootstrap procedure (e.g. Politis and Romano, 1994; Berkowitz and Kilian, 2000). We randomly draw blocks from the matrix of the original group of the considered four processes prior to discretization. The block length l_b is chosen following Hall et al. (1995) as $n^{1/5}$ (rounded to the nearest integer), where n represents the number of time series observations. In our application this leads to a block length of 5 observations. Hence, the bootstrap proceeds as follows:

1. Randomly draw n_b blocks of length l_b from the original matrix of time series;
2. Concatenate those blocks until the length n of the original time series is reached; if n is not a multiple of l_b , the last block is of length $n - n_b \cdot l_b$;
3. Estimate GTE according to Eq. (2);
4. Subtract the estimate of the bias to calculate EGTE;
5. Repeat steps 1–4 to derive the distribution of EGTE.

2.2. Data

We analyze the interdependencies between four volatility processes: OVX, the volatility of the crude oil price, the option based volatility index for the S&P 500 stock index (VIX), volatility of the gold price (GVZ), and the Euro/dollar exchange rate volatility (EVZ). All series are calculated and disseminated by the Chicago Board Options Exchange. The time period considered is June 3, 2008 to May 31, 2017 on a daily basis. Hence each series contains 2303 daily observations. The data is obtained from Datastream and has been downloaded in June 2017. Fig. 1 presents time series plots of the four volatility series. The high volatility that is observed immediately after the global financial crisis and the subsequent euro crisis is followed by more tranquil periods. Visual inspection suggests that it is important to analyze a sufficiently long post-crisis time series to rule out that the results are purely driven by the crisis period and the high volatility therein.

Transfer entropy estimation requires the data series to be stationary. Volatility indices, however, tend to contain a unit root. In addition, Fig. 1 suggests that at least the OVX might have seen a structural break around the end of 2014. Also, our sample period

³ These conclusions are based on a simulation study which is not reported in detail. Results are available on request.

Table 1

Preliminary tests. The table presents test results (test statistics) for the unit root test of Lee and Strazicich (2003), accounting for two possible structural breaks as well as for the Box-Pierce test for first and second order autocorrelation. The 5% critical value of the LS-test is -5.59 .

	LS-test		Box-Pierce test	
	Model 1	Model 2	Lag 1	Lag 2
VIX	−4.24779	−5.46650	0.00040	0.00035
OVX	−3.21151	−4.43955	< 0.00001	< 0.00001
GVZ	−3.91908	−4.70096	0.01442	< 0.00001
EVZ	−3.29099	−4.25029	0.00623	0.00136

includes the global financial crisis which might have induced a structural change in the volatility. Therefore, we use the unit root test of Lee and Strazicich (2003) as implemented by Lips (2017) which can accommodate two structural breaks of which the exact date is unknown. The null hypothesis is that the data contain a unit root. Table 1 holds in columns 2 and 3 the corresponding test statistics for the two considered models. These are: M1 which allows for a structural break in the intercept only, and M2 which allows for a structural break in the intercept and trend. The 5% critical value is -5.59 . Hence, the test results imply nonstationarity of the data in levels at the 5% significance level for all volatility indices. The (not reported) results for the time series in first differences imply stationarity for all series. We therefore consider changes in volatility rather than the level volatility series in our subsequent analyses. Columns 4 and 5 in Table 1 show p -values corresponding to a test for first and second order autocorrelation. The null hypothesis is that there is no autocorrelation at the first and second lag. The results show only marginal autocorrelation which leads us to consider only one lag in the estimation of the GTE.

We also conduct tests for cointegration and no evidence for a common stochastic trend is found. Since there is no theoretical reasoning for a cointegration relationship between these volatility series, we will not consider an error correction term in the VAR model analyzed below.

3. Empirical applications

3.1. Information beyond the VAR model

Previous studies often rely on VAR models to determine the interdependencies of economic and financial variables, using the tools of impulse response functions and forecast error variance decompositions to measure how and to which extent different variables affect each other (cp. Sari et al., 2011; Badshah et al., 2013). While the VAR approach is limited to the estimation of linear dependencies, the transfer entropy approach can account for linear as well as non-linear dependence. To emphasize the importance of such a more general measure, we first estimate a VAR model and subsequently use the residuals as input for transfer entropy calculations. That way we can distinguish between the effects found by the linear VAR approach and additional non-linear dependencies detected by transfer entropy. The VAR model in reduced form is given by

$$\Delta v_t = c + \sum_{j=1}^k \Gamma_j \Delta v_{t-j} + \varepsilon_t, \quad (4)$$

where Δv_t is a (4×1) vector that contains the log-differences of the four volatility indices (VIX, OVX, GVZ, and EVZ), Γ_j are (4×4) parameter matrices, c is a (4×1) vector of constants. If the VAR is

Table 2

VAR results. The table presents the estimation results from a VAR model for the VIX, OVX, GVZ and EVZ. Analytical p -values are reported in parentheses.

	VIX	OVX	GVZ	EVZ
VIX($t-1$)	−0.07847 (0.00201)	0.05621 (0.00084)	0.01655 (0.39103)	0.04291 (0.00838)
OVX($t-1$)	−0.00739 (0.83411)	−0.15041 (0.00000)	0.04933 (0.06603)	0.04970 (0.02810)
GVZ($t-1$)	−0.00394 (0.89906)	0.00428 (0.83504)	−0.09071 (0.00012)	0.00255 (0.89811)
EVZ($t-1$)	0.03506 (0.32878)	0.05291 (0.02617)	0.06161 (0.02401)	−0.09293 (0.00006)
Const	−0.00034 (0.82443)	−0.00018 (0.85627)	−0.00036 (0.75282)	−0.00004 (0.97033)

Table 3

Granger causality tests. The table presents F -statistics for a Granger causality test where the null hypothesis is that the variable used as column head is not Granger caused by the remaining variables out of the set {VIX, OVX, GVZ, EVZ}. p -Values are provided in parentheses.

VIX	OVX	GVZ	EVZ
0.33374 (0.84800)	8.49494 (0.00654)	4.68572 (0.04106)	7.14677 (0.00195)

a suitable model to describe the data, the resulting residual vector, ε_t , is a (4×1) vector of serially uncorrelated innovations. However, if there are nonlinear dependencies between the elements in Δv_t beyond the linear ones described by Eq. (4), the residual series will not be independently distributed.

The VAR is estimated using 1 lag as indicated by the Bayes Information Criterion. Table 2 presents the results. These indicate (at the 5% significance level) an impact of a) past values of the stock market on oil price and exchange rate volatility, b) past changes in oil price volatility impact current exchange rate volatility, and c) past exchange rate changes influence current oil and gold prices. Stock market volatility is not affected by any of the other variables. The estimates of the own lagged impact is always statistically significant on any conventional significance level. The hypothesis that the constant is zero is not rejected which indicates that the level time series – as expected – do not follow a deterministic trend.

In the context of VAR analysis, dependence is regularly analyzed using either impulse response analysis or Granger causality tests.⁴ The impulse response analysis⁵ shows that the interdependencies are rather weak: (orthogonal) shocks to any of the variables die out very fast, at the latest after one period. This is not surprising given the weak dependence structure documented in Table 2.

The presence of Granger causality can directly be inferred using the p -values presented in Table 2. Hence, OVX is Granger caused by VIX and EVZ. VIX is not Granger caused by OVX, GVZ, or EVZ. GVZ is Granger caused by OVX and EVZ, and lastly EVZ is Granger caused by OVX and EVZ. Testing whether the group of three markets Granger causes the respective remaining markets corresponds to an F -test for joint significance of the Granger causal variables (using a robust variance-covariance matrix). The results of this test are summarized in Table 3.

Overall, the results obtained in the VAR analysis are mostly in line with the study of Liu et al. (2013), who document that oil market

⁴ Note that Granger causality and transfer entropy are equivalent in case of normally distributed iid variables (Barnett et al., 2009).

⁵ Detailed results are not reported, plots of the impulse response functions are available in the appendix.

Table 4

Transfer entropy. The table presents the entropy measures based on VAR residuals for information flow between $\hat{\varepsilon}_t^{VIX}$, $\hat{\varepsilon}_t^{OVX}$, $\hat{\varepsilon}_t^{GVZ}$, and $\hat{\varepsilon}_t^{EVZ}$. Information flow is documented as flow towards the named variable from the three remaining variables. Standard errors and *p*-values are based on a time series block bootstrap using 400 replications.

	$\hat{\varepsilon}_t^{VIX}$	$\hat{\varepsilon}_t^{OVX}$	$\hat{\varepsilon}_t^{GVZ}$	$\hat{\varepsilon}_t^{EVZ}$
EGTE	0.01999	0.01883	0.01117	0.01365
se(EGTE)	0.00607	0.00641	0.00591	0.00603
tstat	3.29497	2.93602	1.88801	2.26148
<i>p</i> -Value	0.00098	0.00332	0.05902	0.02373

Table 5

Transfer entropy. The table presents the effective group entropy measures for VIX, OVZ, GVZ, and EVZ. Information flow is documented as flow towards the named variable from the three remaining variables. Standard errors and *p*-values are based on a time series block bootstrap using 400 replications.

	Information flow towards			
	VIX	OVX	GVZ	EVZ
EGTE	0.01957	0.02006	0.01135	0.01273
se(EGTE)	0.00625	0.00669	0.00663	0.00601
tstat	3.13332	2.99747	1.71277	2.11947
<i>p</i> -Value	0.00173	0.00272	0.08676	0.03405

volatility is influenced by lagged changes in the other market volatilities, although the influence is shown to be rather short-lived. With respect to stock market volatility as measured by the VIX, Liu et al. (2013) conclude that the changes in the VIX lead changes in the oil, gold and exchange rate volatility, which corresponds to our findings. However, these results partly contradict the findings by Maghyereh et al. (2017) who do not detect significant volatility spillovers from the equity to the crude oil and gold markets. The VAR-GARCH approach by Arouri et al. (2011) finds significant spillovers between oil and stock market volatility for three out of six countries, where the dominant direction runs from the oil to the stock market.

This might be a hint that the detection of the transmission structure depends crucially on the models applied. The main drawback of the standard VAR approach lies in the fact that it only accounts for linear dependencies. To dig into this issue, we estimate the EGTE using the four VAR residual series $\hat{\varepsilon}_t^{VIX}$, $\hat{\varepsilon}_t^{OVX}$, $\hat{\varepsilon}_t^{GVZ}$, $\hat{\varepsilon}_t^{EVZ}$ as input variables. Those constitute filtered series of the economic variables of interest which are purged for linear dependencies. If the VAR model successfully accounts for all information and the complete dependence structure, the residual series should be purely white noise and EGTE should be zero in all cases.

This, however, is not what we find. The residual based EGTE measures are presented in Table 4. The results are based on a discretization of the data into three bins according to Eq. (3) by choosing the 10 and 90% quantile (a choice used in previous studies such as Dimpfl and Peter, 2013; Sandoval Junior et al., 2015). Effective group transfer entropy and corresponding bootstrapped standard errors are calculated as outlined in the previous section using 400 replications.

The results are in contrast to the VAR results which suggest only weak linear dependencies. EGTE estimates are statistically significant at the 5% significance level for three out of the four markets. We detect significant volatility spillovers that have not been captured by the VAR model, potentially due to their nonlinear nature. The largest and most highly significant spillovers are detected towards the stock and oil market. This suggests that all dynamics between the VIX and the remaining group is in the non-linear domain. Considering the OVZ, the results imply that the sensitivity of the oil market with respect to the other markets is underestimated using the standard

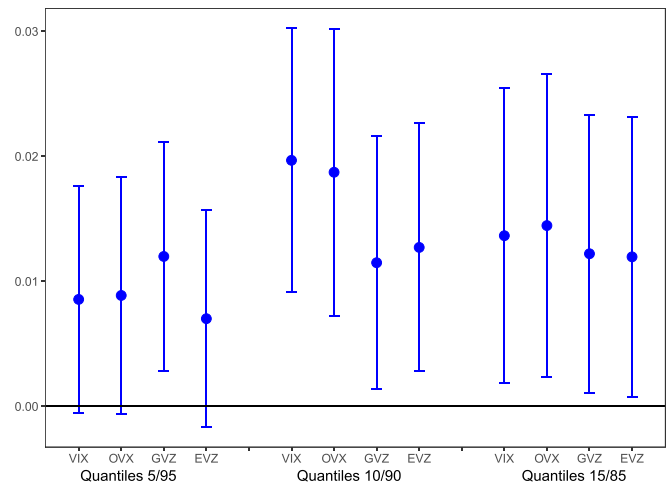


Fig. 2. Transfer entropy results. The figure presents the effective group transfer entropy estimates for VIX, OVZ, GVZ, and EVZ together with their confidence intervals for three different choices of quantile pairs for the symbolic encoding, namely {5, 95} (left), {10, 90} (center), and {15, 85} (right).

VAR approach, as we find significant EGTE estimates using the VAR residuals.⁶

3.2. Direct volatility transmission

The results of the previous section have shown that effective group transfer entropy is able to capture dynamics beyond the standard VAR approach and therefore constitutes an interesting approach to detect volatility spillovers between different markets. In order to measure the full volatility transmission, we apply the GTE approach directly to the changes of the four volatility indices. As before, we assign the data into three bins according to Eq. (3) by choosing the 10 and 90% quantile and calculate effective group transfer entropy and corresponding bootstrapped standard errors. Table 5 presents the results.

All volatility indices are found to be affected by the group of remaining indices. Also, all EGTE estimates are statistically significant, albeit at different significance levels: EGTEs of OVX and VIX are statistically significant at all common significance levels, EGTE of EVZ is significant on a 5% significance level while EGTE of GVZ is only significant at a 10% level.

The highest transfer entropy values are found for OVX and VIX (in the range of 0.02). Their order of magnitude is about twice as high as for GVZ and EVZ. This result shows that the stock market and the oil market are the two most susceptible markets with respect to volatility spillovers from the group of markets. This result is in line with existing literature that documents, in particular, that oil market volatility affects stock market volatility (cp. Arouri et al., 2011; Basta and Molnar, 2018).

To establish robustness of our results with respect to the choice of the quantiles during the discretization, we vary q_1 and q_2 , using the pairs {5, 95} and {15, 85} in addition to {10, 90}. Fig. 2 graphically illustrates the results, presenting point estimates of EGTE and a 90% confidence interval based on bootstrap standard errors. The graphic shows that the dependence is less pronounced when moving further into the tails of the distribution, i.e. using the 5% and 95% quantiles for encoding. However, we have to keep in mind that moving into the tails decisively reduces the number of observations encoded

⁶ Concerning gold market volatility, EGTE estimates based on VAR residuals do not show highly significant spillovers from the other markets to the gold market.

Table 6

Forecast evaluation: RMSPE. The table presents the RMSPE for a 200 days rolling forecast evaluation. The forecasts are based on a naïve model which takes the last observation as the forecast (column 2), on a VAR(1) model (column 3), and linear forecast models whose predictions are averaged using equal weights (column 4) or GTE-based weights (column 5).

	Naïve	VAR(1)	Equal	GTE
VIX	10.38066	7.28434	7.28224	7.28216
OVX	6.72124	4.90771	4.88301	4.88294
GVZ	7.70213	5.42491	5.41509	5.41630
EVZ	7.87278	5.58944	5.58000	5.57911

with A and C, and, thus, also reduces the likelihood of observing a joint tail event which makes the calculation of the frequencies less reliable. When moving towards the center, i.e. using the 15% and 85% quantiles, we observe that the predictability of VIX and OVX is slightly weakened, but still statistically significant. The results for GVZ and EVZ are similar to those obtained with {10, 90} encoding.

4. GTE-weighted forecast averaging

We now turn to a practical application of our GTE measure. In the previous section, we have used GTE to identify whether groups of variables help to predict another variable. Based on this information, we propose to implement a forecasting strategy that uses multiple forecasts weighted by GTE to construct a single point forecast.

More precisely, we will predict a variable X based on all possible subsets u_i which contain two elements of the set $U = \{I, J, Y\}$. Hence, $i = 1, 2, 3$. For each subset u_i , we predict the future outcome of X based on a linear prediction model. The parameters are estimated via OLS in a regression using one lag as

$$x_t = \alpha^i + \beta_1^i x_{t-1} + \beta_2^i u_{1,t-1}^i + \beta_3^i u_{2,t-1}^i + \varepsilon_t^i$$

where u_1^i and u_2^i are the elements contained in u_i . We then iterate forward on this equation to obtain a one-step out-of-sample prediction $x_{i,t+1}^\star$. As U contains three variables in our case, we obtain three predictions $x_{i,t+1}^\star$ based on the three possible subsets u_i .

The predictions are then combined using the estimated GTE⁷ as in Eq. (2). We obtain three GTE measures $\text{GTE}_{u_i \rightarrow X}^i$ by combining the predicted variable X with the predictors in the subsets u_i . As a higher GTE indicates a higher information content for predicting X , we calculate the weights for the forecast averaging as a proportion of the summed GTEs as

$$w_{i,t} = \text{GTE}_t^i / \sum_{i=1}^3 \text{GTE}_t^i.$$

The weights are time dependent as we propose to re-estimate GTE each time anew as the forecast moves through time. The unique prediction of X is then obtained as a weighted average of the $x_{i,t+1}^\star$ as

$$x_{t+1}^\star = \sum_{i=1}^3 w_{i,t} x_{i,t+1}^\star.$$

As benchmark models, we use a naïve forecast based on the lagged observation as $x_{t+1}^\star = x_t$, the prediction based on a VAR

model which is specified as in Eq. (4), and a simple forecast weighting scheme with equal weights $\frac{1}{n}$. The latter has often been shown to be equal or even to outperform more sophisticated weighting schemes in different contexts (for example, Clement and Winkler, 1986; Genre et al., 2013).

To evaluate the forecast, we calculate the root mean squared prediction error (RMSPE). The results presented in Table 6 are based on a 200 days rolling forecast where we always use the maximum of available data to estimate the models. GTE is calculated using the 10% and 90% quantile as in Section 3.2.

Compared to the naïve forecast, our GTE based forecast performs very well. The RMSPE reduction is also statistically significant (based on the test of Diebold and Mariano, 1995). However, the other benchmark models also significantly outperform this forecasting strategy, too. Introducing covariates decisively reduces the forecast error. Again, our GTE based forecast performs well when compared with the VAR(1) based forecast. While the reduction of the RMSPE is not statistically significant, it is still important. In particular, the GTE based forecast performs better (albeit not in a statistically significant sense) than the VAR(1) based forecast for all volatility indices considered.

As forecast averaging has been shown to generally lead to better forecasts, the question is whether our proposed methodology performs also better than an equal weighting scheme which often outperforms a sophisticated weighting scheme (cp. Smith and Wallis, 2009, for a possible explanation). In our application, the GTE based forecast turns out to be slightly more accurate than the equally weighted forecast for three out of four volatility indices. Only in the case of GVZ can the forecast not be improved using GTE. The order of magnitude of the gains is small, but in line with other studies like Banerghansa and McCracken (2010) or Rapach and Strauss (2008) who document that the mean of different forecasts is a fierce competitor for any weighting scheme. Overall, however, the forecast based on the GTE performs well and we believe that in the long run it should be worth the moderate increase of computational time to construct the most accurate forecast available.

5. Conclusion

We measure the interdependence of financial time series by effective group transfer entropy. This procedure circumvents the issues involved in using partial transfer entropy (potential bias and the requirement of very large datasets) and provides the possibility to detect non-linear dependencies, enlarging the realm of multivariate time series analysis.

In our empirical application we consider volatility indices of oil (OVX), the US stock market (VIX), gold (GVZ), and the Euro/US Dollar exchange rate (EVZ) in a post-crisis period. We find significant predictability of each variable, when using the group of the remaining variables. We illustrate that a VAR analysis cannot fully capture the dependence structure, and show that indeed the relationships are to the largest extent non-linear. Consequently, empirical analyses relying on linear time series models underestimate the spillover effects. Our results suggest that the oil and the stock market volatilities are the most affected by volatility transmission from the other markets. Finally, we show that our GTE estimates can be used to improve volatility forecasts compared to a naïve forecast.

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⁷ We chose not to use the effective GTE for two reasons. First, the estimate of the bias is also only an estimate and sometimes during the simulation led to negative EGTE values. It is not obvious how to treat them when combining the forecasts. Furthermore, using GTE directly reduces computational time significantly as the bias has to be reestimated for every step as it depends on the length of the sample.

Appendix A

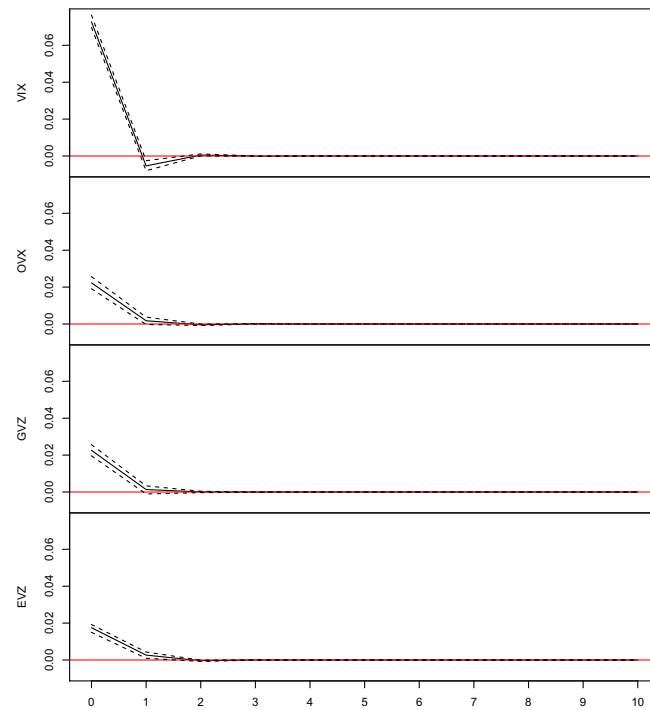


Fig. A.1. Orthogonal impulse response to a shock in VIX. The figure presents the impulse responses of VIX, OVX, GVZ, and EVZ to a one standard deviation shock in the VIX. The orthogonal impulse is calculated using Cholesky decomposition.

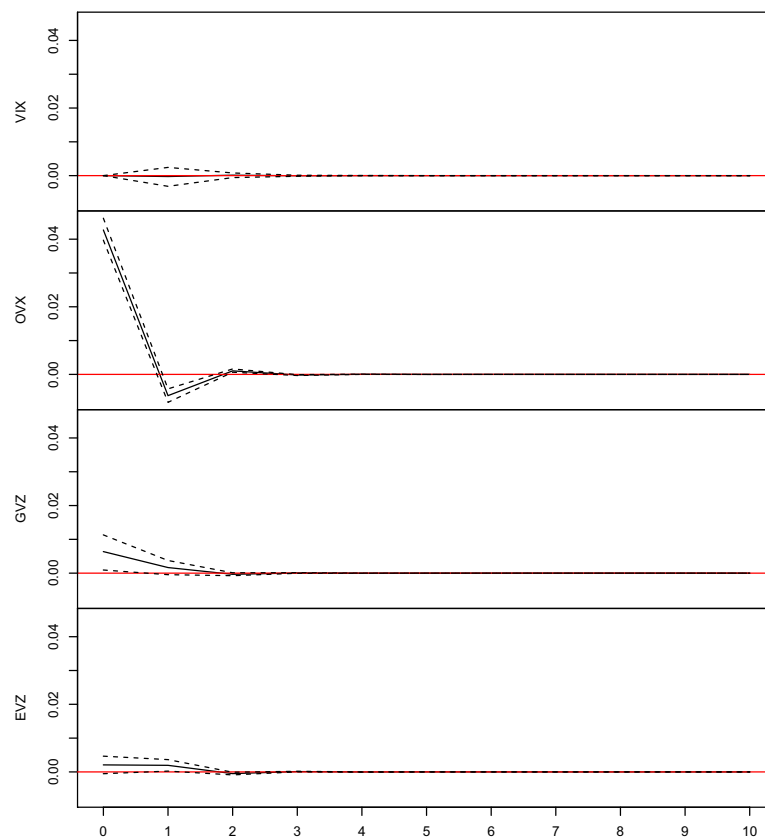


Fig. A.2. Orthogonal impulse response to a shock in OVX. The figure presents the impulse responses of VIX, OVX, GVZ, and EVZ to a one standard deviation shock in the OVX. The orthogonal impulse is calculated using Cholesky decomposition.

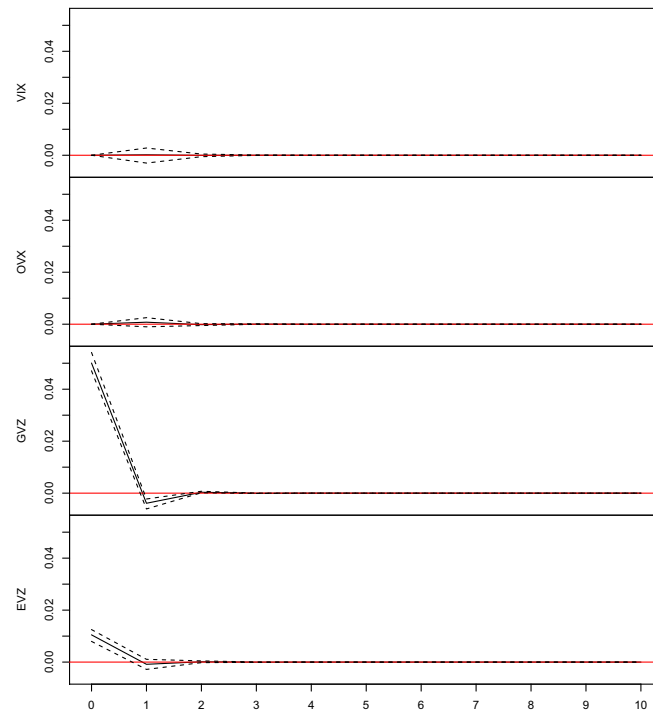


Fig. A.3. Orthogonal impulse response to a shock in GVZ. The figure presents the impulse responses of VIX, OVX, GVZ, and EVZ to a one standard deviation shock in the GVZ. The orthogonal impulse is calculated using Cholesky decomposition.

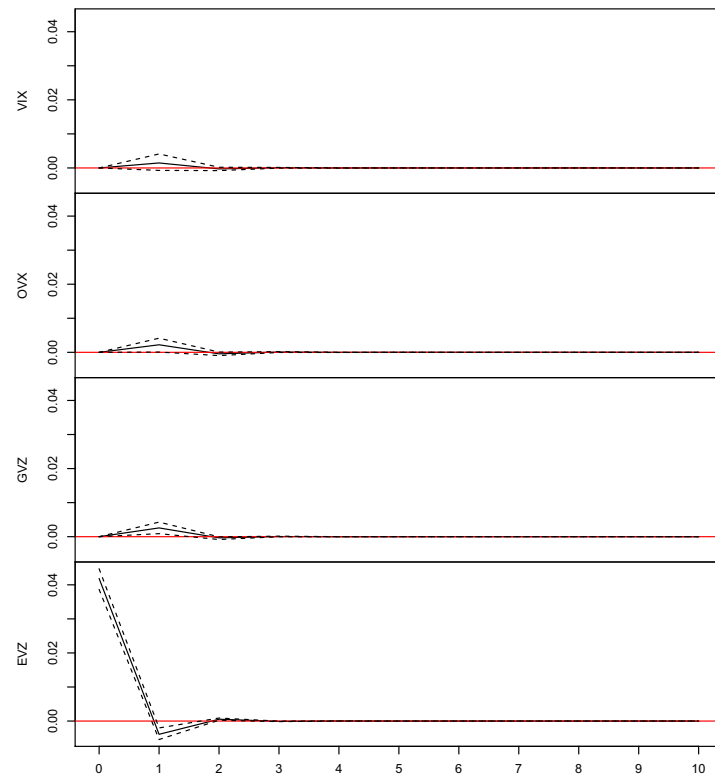


Fig. A.4. Orthogonal impulse response to a shock in EVZ. The figure presents the impulse responses of VIX, OVX, GVZ, and EVZ to a one standard deviation shock in the EVZ. The orthogonal impulse is calculated using Cholesky decomposition.

Appendix B. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eneco.2018.08.008>.

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